

1. A patient arrives at a doctor's office with a sore throat and low-grade fever. After an exam, the doctor decides that the patient has either a bacterial infection or a viral infection or both. The doctor decides that there is a probability of 0.7 that the patient has a bacterial infection and a probability of 0.4 that the person has a viral infection. What is the probability that the patient has both infections?

Solution:

$P(B \cup V) = P(B) + P(V) - P(B \cap V)$. We have $1 = 0.7 + 0.4 - P(B \cap V)$, implying $P(B \cap V) = 0.1$.

2. Consider a pair of events A and B with $P(A) = 0.4$ and $P(B) = 0.7$. Determine the maximum and minimum possible values of $P(A \cap B)$ and the conditions under which each of these values is attained.
3. Prove that for any pair of events A and B , the probability that exactly one of A or B will occur is given by the expression: $P(A) + P(B) - 2P(A \cap B)$.
4. If two balanced dice are rolled, what is the probability that the difference between the two numbers that appear will be less than 3?
5. If four dice are rolled, what is the probability that each of the four numbers that appear will be different?
6. A box contains 100 balls, of which r are red. Suppose that the balls are drawn from the box one at a time, at random, without replacement. Determine (a) the probability that the first ball drawn will be red; (b) the probability that the 50th ball drawn will be red; and (c) the probability that the last ball drawn will be red. You do not have to compute the exact probabilities; finding them in terms of combinations, multiplications, divisions etc. is fine.
7. Suppose that a deck of 52 cards containing 4 aces is shuffled thoroughly and the cards are then distributed among four players so that each player receives 13 cards. What is the probability that each player will receive one ace? You do not have to compute the exact probability; finding it in terms of combinations, multiplications, divisions etc. is fine.

Solution: There are $\binom{52}{4}$ different ways of choosing 4 positions to place the aces in 52 positions $c_1, c_2, \dots, c_{52}$. There are 13 positions to place an ace to the cards $c_1, \dots, c_{13}$ given to the first player, 13 positions to place an ace to the cards $c_{14}, \dots, c_{26}$ and so on. Thus the probability is $13^4 / \binom{52}{4}$.

8. Suppose that two dice were rolled and it was observed that the sum T of the two numbers was odd. What is the probability that T was less than 8?

Solution: The intersection of these two events correspond to T being 3, 5, or 7. Thus the probability of the intersection is $2/36 + 4/36 + 6/36 = 1/3$. The probability that T is odd is $1/2$. Thus the conditional probability is $2/3$.

9. Two boxes contain long bolts and short bolts. Suppose that one box contains 60 long bolts and 40 short bolts, and that the other box contains 10 long bolts and 20 short bolts. Suppose also that one box is selected at random and a bolt is then selected at random from that box. What is the probability that this bolt is long?

Solution: Let B_1 be the event that the first box with 60 long and 40 short bolts is selected and let B_2 be the event that the second box with 10 long and 20 short bolts is selected. Let A be the event that a long bolt is selected. We have

$$P(A) = P(A \cap B_1) + P(A \cap B_2) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) = 1/2 \times 3/5 + 1/2 \times 1/3 = 7/15.$$

10. Suppose that a balanced die is rolled. Let A be the event that an even number is obtained, and let B be the event that one of the numbers 1, 2, 3, or 4 is obtained. Prove or disprove whether the events A and B are independent.

Solution: $P(A) = 1/2$ and $P(B) = 2/3$. $A \cap B$ is the event that either the number 2 or the number 4 is obtained, thus $P(A \cap B) = 1/3$. Since $P(A \cap B) = Pr(A)Pr(B)$ A and B are independent events.

11. Suppose that a machine produces a defective item with probability p where $0 < p < 1$ and produces a non-defective item with probability $1 - p$. Suppose further that six items produced by the machine are selected at random and inspected, and that the results (defective or non-defective) for these six items are independent. What is the probability that exactly two of the six items are defective?

Solution: $\binom{6}{2}p^2(1-p)^4$.

12. Suppose that a machine produces a defective item with probability p where $0 < p < 1$ and produces a non-defective item with probability $1 - p$. Suppose that items produced by the machine are selected at random and inspected one at a time until exactly five defective items have been obtained. What is the probability p_n that exactly n items ($n \geq 5$) must be selected to obtain the five defectives?

Solution: The first $n - 1$ items must be composed of 4 defective and $n - 5$ non-defective, with probability $\binom{n-1}{4}p^4(1-p)^{n-5}$ and the n^{th} item must be defective which has a probability of p . Since these two events are independent $p_n = \binom{n-1}{4}p^5(1-p)^{n-5}$.

13. Three different machines M_1, M_2 , and M_3 were used for producing a large batch of similar manufactured items. Suppose that 20 percent of the items were produced by machine M_1 , 30 percent by machine M_2 , and 50 percent by machine M_3 . Suppose further that 1 percent of the items produced by machine M_1 are defective, that 2 percent of the items produced by machine M_2 are defective, and that 3 percent of the items produced by machine M_3 are defective. Finally, suppose that one item is selected at random from the entire batch and it is found to be defective. What is the probability that this item was produced by machine M_2 ?

Solution: Let B_i be the event that the selected item was produced by machine M_i and let A be the event that the selected item is defective. Note that $P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3) = 0.2 \times 0.01 + 0.3 \times 0.02 + 0.5 \times 0.03 = 0.023$. Furthermore by Bayes' Theorem we have $P(B_2|A) = P(B_2)P(A|B_2)/P(A)$. Therefore $P(B_2|A) = 0.3 \times 0.02/0.023$.

14. Suppose that a baseball hitter has a probability of success $p = 0.7$. What is the probability that she hits (has success) more than 6 times out of a total of 15 throws. You may make use of the following outputs from scipy.stats: $\text{binom.pmf}(5, 15, 0.7) = 0.003$, $\text{binom.cdf}(5, 15, 0.7) = 0.004$, $\text{binom.pmf}(6, 15, 0.7) = 0.012$, $\text{binom.cdf}(6, 15, 0.7) = 0.015$.
15. Suppose that expectation of a random variable X is $E(X) = 5$. What is $E(3x - 5)$?
16. Let X_1, X_2 , and X_3 be independent random variables such that $E(X_i) = 0$ and $E(X_i^2) = 1$ for $i = 1, 2, 3$. What is the value of $E[X_1^2(X_2 - 4X_3)^2]$?

17. Suppose that a random X can take each of the five values $-2, 0, 1, 3, 4$ with equal probability. Compute the variance of X using the formula $Var(X) = E(X^2) - (E(X))^2$. Also compute the variance of $Y = 2X - 10$.

18. Suppose that the random variable X can take only the three values $-1, 0, 1$, and that each of these three values has the same probability. Also, let the random variable Y be defined by the relation $Y = X^2$. Show that X and Y are dependent but uncorrelated.

Solution: Since the value of Y is completely determined by the value of X , the variables are dependent. On the other hand, $E(XY) = E(X^3)$ and since X^3 is the same variable as X , we have $E(X^3) = E(X) = 0$. Since we have $E(XY) = 0$ and $E(X)E(Y) = 0$ we have $Cov(X, Y) = E(XY) - E(X)E(Y) = 0$ and thus the correlation coefficient is 0.

19. Suppose that X and Y are random variables such that $Var(X) = 9, Var(Y) = 4$, and $\rho(X, Y) = -1/6$. Determine $Var(X + Y)$.

Solution: $\sigma(X) = 3, \sigma(Y) = 2$. We have $-1/6 = Cov(X, Y)/(2 \times 3) \Rightarrow Cov(X, Y) = -1$. Since $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$, we have $Var(X + Y) = 9 + 4 - 2 = 11$.

20. Suppose that on the average 5 students show up to my office hours every half an hour. What is the probability that there will be more than 7 students in a particular office hour period consisting of one hour? You may make use of the following outputs from `scipy.stats`: `poisson.cdf(6, 5) = 0.76, poisson.cdf(7, 5) = 0.87, poisson.cdf(6, 10) = 0.13, poisson.cdf(7, 10) = 0.22`

21. Suppose that X has the normal distribution with mean 5 and standard deviation 2. Find out the value of $P(1 < X < 8)$. You may make use of the following outputs from `scipy.stats`: `norm.pdf(2) = 0.054, norm.cdf(2) = 0.977, norm.pdf(3) = 0.004, norm.cdf(1.5) = 0.933`

Solution: $Z = (X - 5)/2$ and we have $P(1 < X < 8) = P((1 - 5)/2 < Z < (8 - 5)/2) = P(-2 < Z < 3/2)$. Note that $P(-2 < Z < 3/2) = P(Z < 3/2) - P(Z \leq -2)$. We have that $P(Z \leq 2) = 0.977$ and thus $P(Z \geq 2) = P(Z \leq -2) = 1 - 0.977 = 0.023$. We also see that $P(Z < 3/2) = P(Z \leq 3/2) = 0.933$. We get the probability as $0.933 - 0.023 = 0.91$.

22. Recall that Chebyshev's inequality states that the probability that any given random variable will differ from its mean by more than k standard deviations cannot exceed $1/k^2$. Suppose that X is a random variable for which the mean is $E(X) = 10$. Furthermore, $P(X \leq 7) = 0.2$, and $Pr(X \geq 13) = 0.3$. Show that $Var(X) \geq 9/2$.

Solution: The probability that X is $3 = k\sigma$ units away from the mean is $0.2 + 0.3 = 0.5$. It should be that $0.5 \leq 1/k^2 = \sigma^2/9$ and therefore $\sigma^2 \geq 9/2$.

23. Suppose we observed n coin flips, where probability p of heads is unknown. We wish to estimate p with the average number of heads in n flips. Compute the bias, variance, and the mean squared error of this estimator.

24. A random sample of n items is to be taken from a distribution with mean μ and standard deviation σ .

- (a) Use the Chebyshev inequality to determine the smallest number of items n that must be taken in order to satisfy the following relation:

$$P(|\bar{X}_n - \mu| \leq \frac{\sigma}{4}) \geq 0.99$$

- (b) Use the central limit theorem to determine the smallest number of items n that must be taken in order to satisfy the relation in part (a) approximately. You may make use

of the following outputs from `scipy.stats`: $norm.ppf(0.995) = 2.58$, $norm.ppf(0.95) = 1.64$, $norm.cdf(0.995) = 0.84$, $norm.cdf(0.95) = 0.83$.

25. We take a random sample of size 64 from a distribution with unknown mean μ that has standard deviation $\sigma = 4$. Assume the sample mean is 6. What is the 90% confidence interval for mean μ ? You may make use of the following outputs from `scipy.stats`: $norm.ppf(0.995) = 2.58$, $norm.ppf(0.95) = 1.64$, $norm.cdf(0.995) = 0.84$, $norm.cdf(0.95) = 0.83$.
26. We take a random sample of size 4 from a distribution with unknown mean μ and unknown σ . The values of the sample are 3, 5, 2, 6. What is the 90% confidence interval for mean μ ? You may make use of the following outputs from `scipy.stats`: $norm.ppf(0.95) = 1.64$, $t.ppf(0.95, 3) = 2.35$, $norm.ppf(0.995) = 2.58$, $t.ppf(0.995, 3) = 5.84$.
27. Suppose that the lengths in millimeters of metal fibers produced by a certain process have the normal distribution with unknown mean μ and unknown variance σ^2 , and the following hypotheses are to be tested: $H_0 : \mu \leq 5.2$ and $H_A : \mu > 5.2$. Suppose that the lengths of 15 fibers selected at random are measured, and it is found that the sample mean $\bar{X}_n$ is 5.4 and sample standard deviation $\hat{\sigma}_1 = 0.4226$. Find out whether to reject H_0 at significance 0.05. You may make use of the following outputs from `scipy.stats`: $t.ppf(0.95, 14) = 1.76$, $norm.ppf(0.95) = 1.64$, $t.ppf(0.995, 14) = 2.98$, $norm.ppf(0.995) = 2.58$.
28. Find out the p value of the hypotheses above. You may make use of the following output from `scipy.stats`: $t.cdf(1.833, 14) = 0.956$.
29. Associate each plot below with one of the correlation coefficient values: 1, 0.8, 0.4, 0, -0.4, -0.8, -1.



Figure 1: Points (x, y) .

30. Discuss briefly what the phrase *correlation does not imply causation* means and provide an example.
31. Discuss what Z-score normalization is and why it is useful in handling data compatibility issues while doing data cleaning.
32. The exponential distribution has cdf $1 - e^{-\lambda x}$. Describe how we could sample two points from this distribution using inverse transform sampling.
Solution: We first find the inverse of $F(x) = 1 - e^{-\lambda x}$. We have $e^{-\lambda x} = 1 - F(x)$ which implies $-\lambda x = \ln(1 - F(x))$ and $x = -\ln(1 - F(x))/\lambda$. We sample $u = F(x)$ from $[0, 1]$ uniformly random, say 0.2 and we plug it in the equation to get our first sample from the distribution: $x = -\ln(1 - 0.2)/\lambda$. Doing this once more, say with 0.4 from uniform random sampling we have a second sample from the distribution: $x = -\ln(1 - 0.4)/\lambda$.
33. Describe why we need multiple testing correction and a method to resolve it.